

BRAC University
Department of Computer Science and Engineering

Examination: Quiz 01
Duration: 20 Minutes

Semester: Summer 26
Full Marks: 15

CSE420: Compiler Design

Name:	ID:	Section:
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1. Why is ambiguity a problem in compilers? Can ambiguous grammars ever be transformed into unambiguous ones? Justify your answer. (3)

2. Show how a bottom up parser with the given grammar rules will process the following input string. digit - id + (digit) (12)

$E \rightarrow E + T \mid E - T \mid E * T \mid E / T \mid T$

$T \rightarrow F \mid C$

$F \rightarrow (E) \mid \text{digit} \mid \text{id}$

$C \rightarrow \text{digit} - \text{id}$

Stack	Input Buffer	Action

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1. You are writing a FLEX program for a language with identifiers, keywords, integer literals, and assignment operators, given as follows:

$[a-zA-Z][a-zA-Z0-9]^* \rightarrow \text{ID}$

$\text{for|while} \rightarrow \text{KEYWORD}$

$[0-9]^+ \rightarrow \text{NUM}$

$= \rightarrow \text{ASSIGN}$

Explain, using the code above, how token definition order influences parsing correctness. (3)

2. Show how a bottom up parser with the given grammar rules will process the following input string. (digit / (digit)) (12)

$E \rightarrow E + T \mid E - T \mid E * T \mid E / T \mid T$

$T \rightarrow F \mid C$

$F \rightarrow (E) \mid \text{digit} \mid \text{id}$

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1. In a C-like compiler pipeline, why does the lexical analyzer ‘discard’ (or never tokenize) a line such as **#include <stdio.h>** ? (3)

2. Show how a bottom up parser with the given grammar rules will process the following input string. id + digit / (id) (12)

$E \rightarrow E + T \mid E - T \mid E * T \mid E / T \mid T$

$T \rightarrow F \mid C$

$F \rightarrow (E) \mid \text{digit} \mid \text{id}$

$C \rightarrow \text{digit} - \text{id}$

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